

Ultrafast X-Ray Experiments

**Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21**

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1. Introduction

The experiment EXP21 - Ultrafast X-ray experiments is designed to provide basic hands-on knowledge about the methods of ultrafast pump-probe spectroscopy as well as X-ray emission spectroscopy and related X-ray techniques. This is done through the usage of the virtual environment VLab - a software that simulates a beamline of the European XFEL. The following procedures are part of the experiment: characterising the optics of the X-ray beam used, making the preparatory estimates and calculations surrounding the setup of the experiment and finally executing a pump-probe measurement of the relaxation process of the iron-compound $[\text{Fe}(\text{bipy})_3]^{2+}$ at different time delays.

2. Theoretical Background

2.1. Synchrotron radiation

When accelerating electrons or other charged particles to relativistic speeds along a curved trajectory they emit so-called synchrotron radiation. It gets its name from the ring accelerators, the synchrotrons, where it was first observed. A typical synchrotron facility consists of a linear accelerator leading into a booster ring in which the electrons are brought up to relativistic speeds [1][2]. The electrons are then fed into a storage ring in which they continue to circulate at relativistic speeds. From the storage ring the beam-lines lead tangentially away in which the synchrotron radiation is utilized in experimental hutches [1][2].

Synchrotron radiation is bright X-radiation with a wide tunability in energy, which is useful for examining structures and dynamics of materials, as X-ray photons have energies in the order of several kiloelectronvolts (keV) [2]. This means their wavelengths are in the order of magnitude of Angstroms and comparable to the spacing of atoms in solid materials. This is useful, as it means that atoms in a crystalline array can diffract X-ray beams and the interference patterns can be measured [1].

There are four generations of synchrotrons with the main difference being the brilliance achieved [1]. The first was the bending magnet in the synchrotron as it was used for high energy collider physics with no additional modifications. The second generation introduced the wiggler, a configuration of dipole magnets with alternating polarity to achieve synchrotron radiation along linear path. The third generation synchrotron radiation facilities were designed for low emittance of the beams with more effective linear magnet configurations, the undulators [2]. A low emittance means a smaller angular divergence of the beam and higher brilliance [2].

2.1.1. Free Electron Laser *FEL*

Free-electron lasers are coherent radiation sources that are capable of even higher brilliance than third generation synchrotron facilities [3]. They are, in theory, able to operate at any arbitrary wavelength or spectrum of wavelengths with the limit being the energy and quality of the accelerated electron beam [3]. FELs use a process called self-amplified stimulated emission (SASE). SASE free-electron lasers typically use monochromatic electron beams with a long, high-quality undulator around the order of magnitude of a few hundred metres. The light waves emitted by the electron bunches will line up in phase and amplify themselves through superposition. As the electrons oscillate down the undulator they are also interacting with the magnetic components of the light they themselves are generating. The transverse component of the electrons oscillation and the magnetic field component of the emitted light create a Lorentz force which acts upon the electrons and moves them along axial directions. This leads to the electrons creating microbunches inside their conventional bunches [1].

The duration of a single X-ray pulse can also be reduced to even less than a few femtoseconds. This allows for an investigation of molecular processes and photochemical reactions through a form of sequential snapshots with a characteristic time of under a picosecond [1].

2.2. Pump-probe-spectroscopy

The principle of pump-probe-spectroscopy is directing two lasers at the same sample with a certain time delay so that the first laser, an optical laser, can excite the system in some way and the second laser, an X-ray pulse can measure the response (see Figure 1). In a FEL such as the European XFEL this can be done with ultrafast X-ray pulses on a timescale of a few femtoseconds (10^{-15} s). Varying the time delay allows for a series of sequential snapshots of the process observed and varying the wavelength of the excitation laser allows for an investigation of different processes [4].

The selection of the different discrete time delays is important. For negative time delays ($\Delta t < 0$) the probe pulse arrives before the pump pulse and thus measures just the ground state and for a delay of $\Delta t = 0$ both pulses overlap exactly, whereas for positive delays ($\Delta t > 0$) the probe pulse measures the developing dynamics.

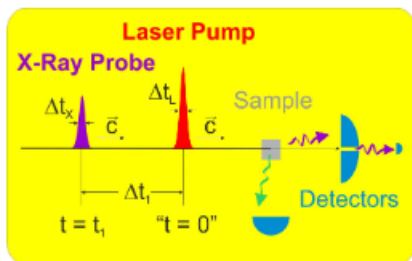


Figure 1: Schematic principle of the pump-probe experiment: the optical pump pulse excites the sample and the delayed X-ray probe pulse measures its response. Varying the delay between both pulses provides time-resolved snapshots of the relaxation process [4].

2.2.1. The pump-pulse

The pump pulse acts as a kick-off signal for the process. It excites the molecule at a well-defined time ($t = 0$) from its thermal ground state into a higher energetic state [4].

In this experiment specifically the pump pulse is an ultra-short optical laser-pulse with a wavelength in the VIS/UV range which excites the iron-compound $[\text{Fe}(\text{bipy})_3]^{2+}$ from its singlet low-spin-state (1A_1) into the excited state $^1\text{MLCT}$ (metal-to-ligand charge transfer) after which the relaxation process starts [4].

2.2.2. The probe-pulse

The probe pulse is a second pulse that hits the sample with a known time delay to 'photograph' the current state. In this experiment an ultra-short X-ray pulse from the European XFEL is used. It ionises the 1s core electron of the central iron atom. The resulting X-ray emission ($K\beta$ -XES) is measured which is sensitive to spin- and structure changes of the iron [4].

2.2.3. Transient difference spectrum

Most of the time during the experiment, only a fraction of the molecules in the probe are in an excited state. This means the pure excited spectrum is hard to see directly. To get a useful reading of the data the transient signal in the difference spectrum is used:

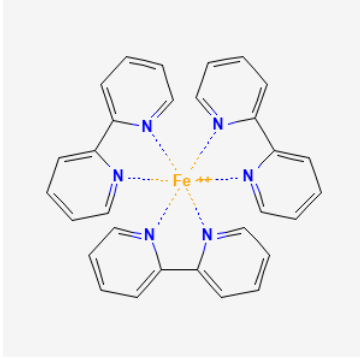
$$\text{transient signal} = I_{\text{laser on}}(\Delta t) - I_{\text{laser off}} \quad (1)$$

The transient signal shows the physical changes the best: Negative parts of the signal mean a depopulation - places in the spectrum where the intensity lessens as the ground state is emptied because of the laser, positive parts are new spectral features such as the growing of a spin-sensitive satellite peak which indicates the existence of intermediate states.

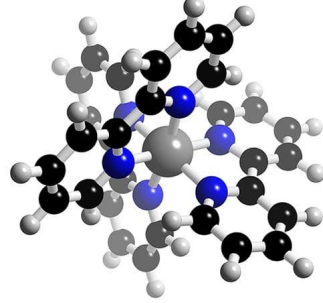
When plotting the intensity of these notable energies against the time delays, you get so-called kinetic traces from which the lifespans of the intermediate states is able to be determined.

2.3. The investigated molecule

We investigated a molecule formed of an Iron center (Fe^{2+}), a transition metal ion, and 3 attached ligands (bipyridine). This forms a chemical complex compound. The central atom acts as an electron acceptor, while the ligands act as the electron donor.



(a) Two-dimensional chemical structure of the iron complex, showing the three bidentate bipyridine ligands around the iron centre [PubChem].



(b) Three-dimensional representation of $[\text{Fe}(\text{bipy})_3]^{2+}$, depicting its approximately octahedral coordination geometry [ChemTube3D].

Figure 2: Two- and three-dimensional views of the investigated $[\text{Fe}(\text{bipy})_3]^{2+}$ complex. Together, they show both the connectivity of the ligands and the spatial arrangement around the iron centre.

This bonding can take place because of the d^6 electron configuration of Fe^{2+} and the nitrogen atoms in the pyridine rings. In our case we have two bridged pyridine rings that form the uncharged bidentate ligand bipyridine (abbreviated bipy). Both nitrogen atoms in each of the 3 ligands bind to the central Fe^{2+} atom.

2.3.1. Electronic Structure of $[\text{Fe}(\text{bipy})_3]^{2+}$

Because of the electrostatic interaction between the ligands and the d electrons of the Fe^{2+} , the 5 d orbitals, that normally would be energetically equivalent, with a maximised spin multiplicity, are now energetically raised or lowered due to the nature of interaction with the ligand field (see Figure 3). The energy of d orbitals pointing to the ligands (e_g) is raised, while the energy of those pointing between the ligands is lowered (t_{2g}). The difference of energy Δ_{oct} is called the strength of the crystal field.

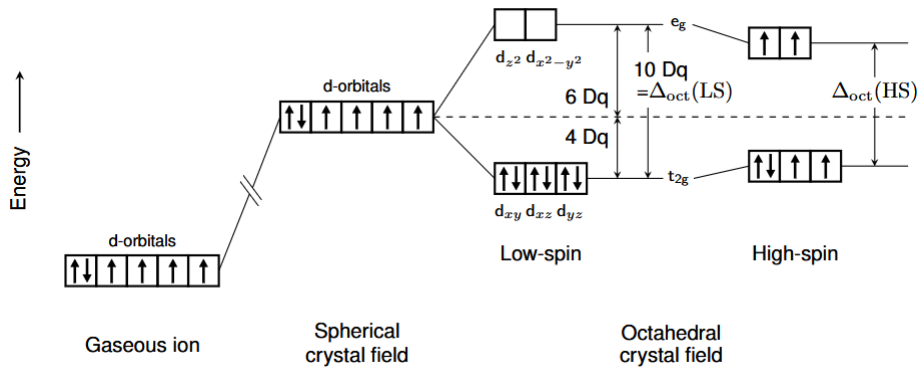


Figure 3: Splitting of the five d orbitals of a d^6 transition-metal ion in an octahedral ligand field. The field separates the higher-energy e_g orbitals from the lower-energy t_{2g} orbitals and thereby determines whether low-spin or high-spin configurations are favoured [4].

Following Hund's rule the lower-energy orbitals (t_{2g}) are filled first (one by one with electrons of the same spin). In the low-spin state our d^6 metal ion has no unpaired electrons (total Spin $S = 0$). That represents the singlet state 1A_1 . In the high spin state it has the maximum of unpaired electrons ($S = 2$).

We have a quintet state 5T_2 . The energy difference Δ_{oct} is smaller for HS configurations, because the e_g level is filled and those electrons can push the ligand atoms further away than in LS configurations. Excitation to the HS state occurs in this experimental setup through photoexcitation to higher lying MLCT bands followed by relaxation.

2.3.2. Relaxation process

The actual process we want to look at is the relaxation process that occurs after exciting the iron compound from the singlet state 1A_1 into the MLCT (metal to ligand charge transfer) band. In the electronic ground state $[\text{Fe}(\text{bipy})_3]^{2+}$ is a low spin complex while it's a high spin complex when excited to the MLCT band.

There are two different ways the deactivation could occur. It can occur either directly to the high spin state $^5T_{2g}$ or indirectly via the transient state $^3T_{1g}$. Both processes are equally possible. The average lifetime of the MLCT band is $\tau_{MLCT} \approx 100$ fs while the average lifetimes of the transient state $^3T_{1g}$ and the high spin state $^5T_{2g}$ are $\tau_3 < 100$ fs and $\tau_5 \approx 1$ ns respectively (see Figure 4) [4].

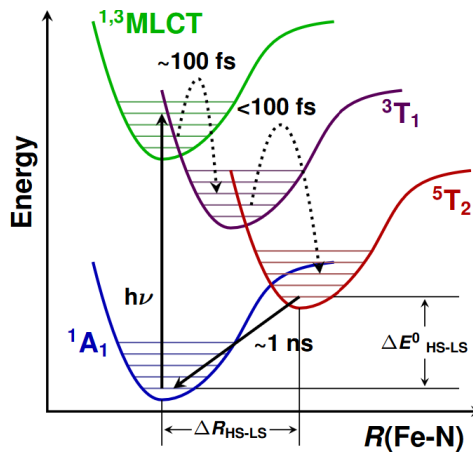


Figure 4: Potential-energy curves describing the excited-state dynamics of $[\text{Fe}(\text{bipy})_3]^{2+}$ in water after excitation into the MLCT band. The diagram demonstrates the suggested relaxation pathways through a transient triplet state and towards the long-lived quintet state [4].

3. Experimental setup

The entire experiment is conducted in VLab, a software modeled after a hutch in the experimental hall of the European XFEL in Schenefeld. The European XFEL is 3.4 km long and runs from the DESY research center to the European XFEL campus. A 1.9 km long superconducting linear accelerator feeds electrons with energies up to 17.5 GeV into a branched tunnel system with two 212 m long undulators that generate hard X-radiation in the range of 5 keV to 25 keV and one 125 m long undulator that generates soft X-radiation in the range of 0.25 keV to 3 keV.

VLab recreates the Femtosecond X-ray Experiment research platform FXE. The simulation has no built-in structural errors, but there are statistical errors when it comes to the photon count.

3.1. Beamline

After the SASE process in the undulators, the beam is neither monochromatic nor really polychromatic - it's a so-called "pink" beam - which is why it goes through a monochromator before entering any of the optical elements on the optical bench in the experimental hutch.

The beamline in VLab uses a two-crystal monochromator (see Figure 5) which uses the diffraction in the first crystal to absorb all wavelengths that do not interfere constructively [4].

Before entering the optical bench there's also a beam splitter built with a linear diamond grating

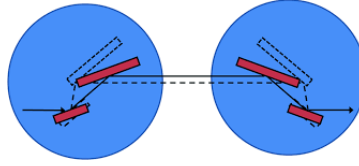


Figure 5: Optical layout of a two-crystal monochromator used to reduce the bandwidth of the X-ray beam. The crystal arrangement selects a narrow energy range through successive Bragg reflections [5].

which can be inserted, if so desired, to look at the ± 1 order diffractions. The beam splitter is built in such a way, that one of the first order diffractions is sent to a spectrum analyser which consists of a bent crystal and a detector [4].

3.2. Focusing optics

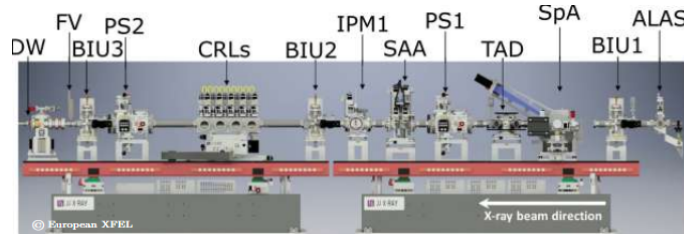


Figure 6: Overview of the FXE optical bench and its diagnostic and focusing components: BIU 1,2,3 (beam imaging units), SpA (spectrum analyser), TAD (time arrival detector), PS 1,2 (power slits), SAA (solid attenuator assembly), IPM1 (intensity position monitor), CRLs (compound refractive lenses) [4].

In VLab as well as in real life, there are multiple optical elements for controlling and characterising the beam (see Figure 6). Listed here are the ones we actively used during this experiment.

There are three BIUs which measure the intensity distribution of the beam perpendicular to its propagation direction. In VLab these intensity distributions can be saved as files with a size of 420x420 pixel and the screen dimension 10 mm x10 mm leading to a resolution of 0.024 mm/px.

The power slits are there to restrict the areas of the beam, for this experiment they are fully opened.

As some components and detectors of the beamline will be damaged when hit with the full force of the X-ray beam (such as the BIUs), it is necessary to have a way of weakening the beam without changing anything else about its optics. This is done with the solid attenuators of the SAA which are able to weaken the beam by multiple orders of magnitude. It consists of an assembly of four arms (or rods) which each have a selection of foils of varying thicknesses to hold into the beam. The foils are made of diamond (C^*), silicon (Si) and boron carbide (B_4C) [4].

To focus the beam, compound refractive lenses made of beryllium are used. X-ray optics need a different approach to lenses than UV/VIS light as can be seen when looking at the dimensionless wavelength dependent index of refraction: $n(\lambda) = n_R(\lambda) + in_I(\lambda)$. The real part n_R is the refractive index and measures the phase velocity of light. For wavelengths in the X-ray regime it drops to less than unity, which means the refractive power is extremely low. This means focusing of the beam has to be performed with concave lenses rather than convex lenses one would use to focus an optical beam. The lenses are made of beryllium as it is one of the available materials with a low atomic number which means that the lenses have low X-ray absorption [4].

The focal distance F of single refractive X-ray lenses is dependent on the lens radius R and the refractive decrement δ through $F = \frac{R}{2\delta}$ but as δ is very low it's necessary to use stacks of several lenses. Now the

focal distance of N lens elements is given by:

$$F = \frac{R}{2N\delta} \quad (2)$$

The CRLs in VLab are in the form of ten retractable carrier cassettes which are each loaded with up to ten individual Be lenses with varying R (see Table 2). The carrier cassettes can be placed into the beam at fixed positions which are already aligned to perfectly focus the beam - saving the work of manually finding the right positions that would be required when actually working at the FXE [4].

3.3. Sample environment

The next section of the hutch modeled in VLab is the sample environment in which the actual experiment takes place. It consists of the sample table on which the pump-laser and the X-ray hit the probe and secondary spectrometers as well as the large pixel detector (LPD). For this experiment the liquid beam injector on the sample table is used to make sure that the probe for every pump and probe cycle is fresh [4].

There are two secondary spectrometers available in VLab: the Von Hamos spectrometer and the Johann spectrometer. The Von Hamos is a dispersive spectrometer which means that it is better suited for recording an entire spectrum while the Johann is a monochromatic spectrometer and better suited for accurately measuring the wavelength of single small wavelength ranges of interest. The Von Hamos in VLab consists of up to 16 cylindrically curved crystals [4].

The output of the spectrometers can be measured by two different detectors: the Gotthard detector which is a MHz compatible strip detector or the Jungfrau detector which is a 10Hz compatible pixel detector. These detectors are made out of silicon and work because of the X-ray generating conduction band electrons whose charge quantity is then measured. This means the detectors sensitivity is energy-dependent.

The LPD was developed specifically for the European XFEL. It's an 1 mega pixel detector with the ability to resolve single pulses within a pulse train. To ensure that the directly transmitted X-ray beam doesn't destroy any of the sensors in the detector there's a small hole which allows the beam to pass without directly hitting any of them.

In this section there's also another more precise BIU - the X-ray eye - which also measures intensity distributions with a size of 420x420 pixel and the screen dimension 2 mmx2 mm leading to a resolution of 0.005 mm/px.

3.4. Diagnostic table

The relevant part of this last section of the beamline is the final beam stop, made of a sequential set of B4C and Cu attenuators. The failsafe is a "burn-through" monitor placed in front of the beam stop materials. If the beam is accidentally focussed to the end of the hutch, it will burn a hole in this monitor and generate visible light which switches off the electron photoinjector.

In VLab the worst thing that can happen in regards to radiation safety is that you burn through one of your detector screens and have to start over again which is of course much less punishing than a potentially life threatening scenario in real life.

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.

As the absorption edge for the Fe $K\beta_{1,3}$ line is at 7065 eV, the energy of the X-ray beam is set to 7400 eV, to avoid the ripples near the reaction (as illustrated by the X-ray absorption spectrum of $[\text{Fe}(\text{terpy})_2]^{2+}$ Figure 7) but without too strong of a loss in intensity.

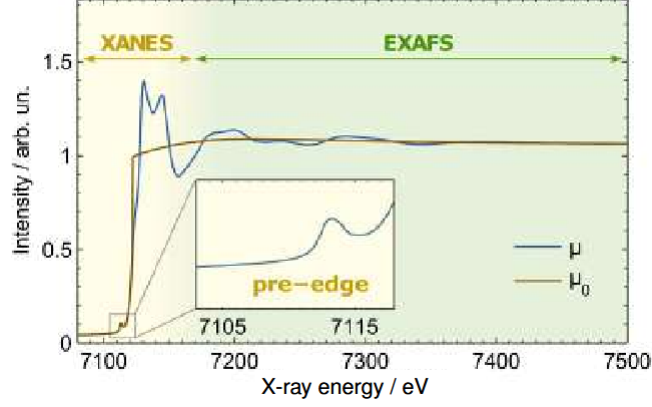


Figure 7: X-ray absorption spectrum of $[\text{Fe}(\text{terpy})_2]^{2+}$ at the Fe K edge, showing the XANES and EXAFS regions together with the background contribution μ_0 . The spectrum illustrates why the beam energy is chosen above the absorption edge, while avoiding the strongest near-edge structure [4].

To measure the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturation of the beam imaging unit 2 (BIU2):

Table 1: Attenuator foils used

Rod	Attenuator foil
1	C 0.1
2	Si 0.05
3	C 0.4
4	B4C 0.8

The size and position of the x-ray is now measured using BIU2. Then the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24° .

For the following experiment the pink X-ray beam is used even though the monochromatic has a higher resolution, as the intensity of the beam suffers about three orders of magnitude under the monochromator (see Figure 8).

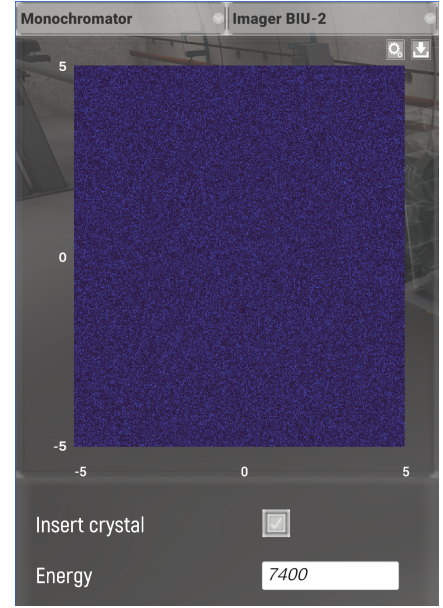
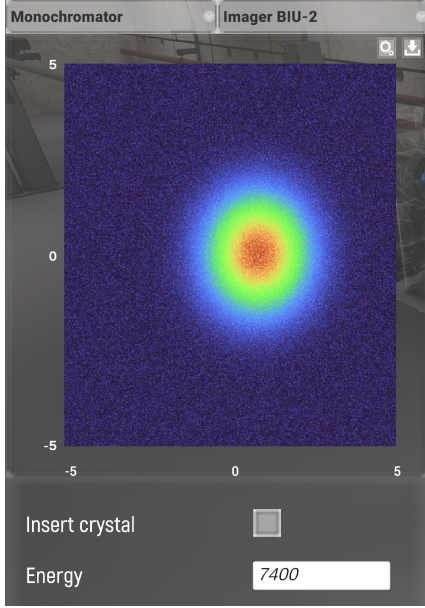
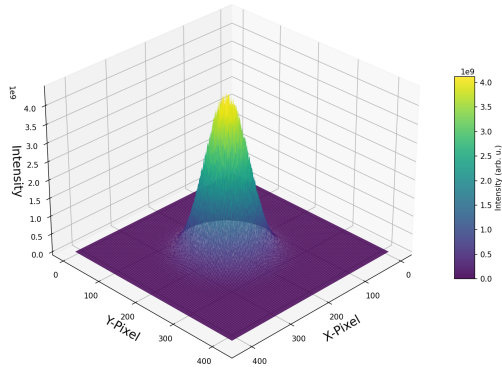


Figure 8: Beam images recorded with the same attenuator settings: the pink beam is shown on the left and the monochromatic beam on the right. The comparison makes the strong intensity loss caused by the monochromator visible, despite the improved spectral selectivity.

3D Reconstructed Pink Beam Intensity Profile



3D Reconstructed Monochrom. Beam Intensity Profile

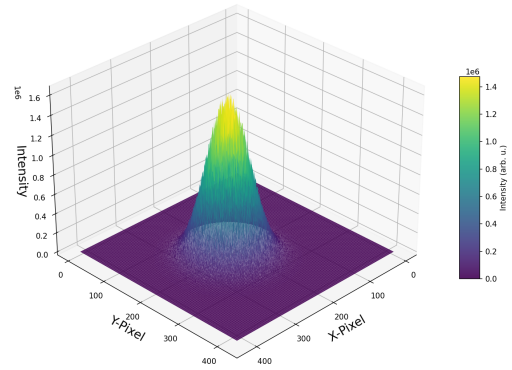


Figure 9: Three-dimensional intensity distributions of the pink beam (left) and the monochromatic beam (right) measured with identical attenuators. The plots show how the monochromator changes the recorded beam intensity while preserving the overall spatial beam profile.

The last step of diagnosing the beam line is using the refractive beryllium X-ray lenses to focus the beam at the imaging unit X-ray eye. It's important to appropriately adjust the attenuator foils as to not burn the imaging unit with a focussed beam.

Table 2: A list of the beryllium lenses available in VLab together with the arms they were mounted on (arm 10 the most downstream), the Number of lenses in each cassette N and the central curvature of the lenses R [4].

Arm	10	9	8	7	6	5	4	3	2	1
N	10	10	8	7	5	9	4	10	6	1
R/mm	0.5	0.5	0.5	0.5	0.5	1.0	0.5	1.5	1.5	0.5

With beryllium lenses 1, 2 and 5 (taken from Table 2) and appropriate attenuator foils inserted, a beam

with the FWHM of $\approx 13 \mu\text{m}$ (siehe Figure 10) is achieved. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [4] and the properties of the lenses themselves taken from Table 2:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$

$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4.76 \text{ m}$$

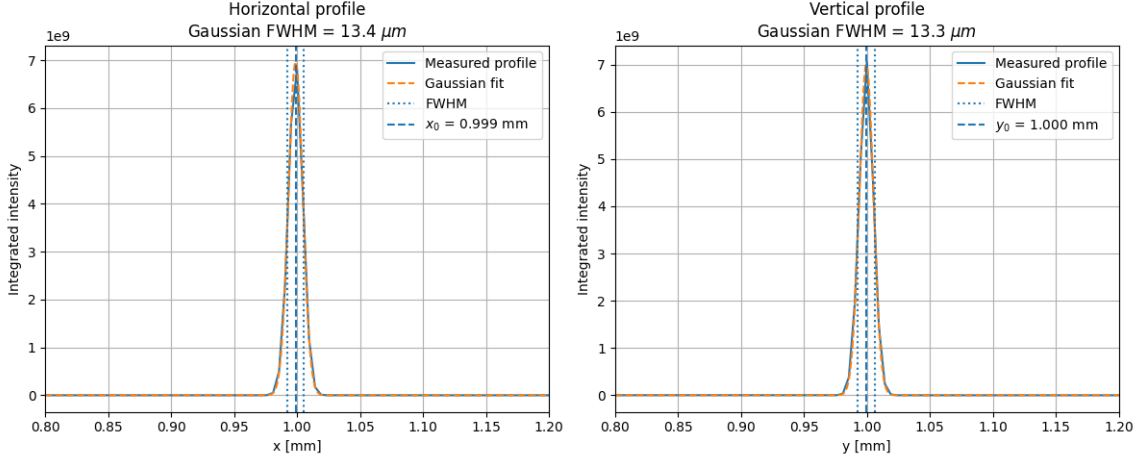


Figure 10: Horizontal and vertical intensity profiles of the focused X-ray beam together with Gaussian fits. The resulting FWHM of approximately $13.5 \mu\text{m}$ quantifies the beam size achieved at the focus.

4.1.2. Preparatory Estimates for the Experiment

The transition metal complex to be studied, $[\text{Fe}(\text{bipy})_3]^{2+}$, is obtained through dissolving the salt $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ in deionized water to the desired concentration. The actual sample is a powder consisting of small microcrystals that contain incorporated water which remained in the crystal lattice after the crystallization process.

A minimum volume of 50 ml is required to operate the liquid jet in the experiment. To ensure that the sample is completely refreshed between probe pulses, the jet needs to have a certain flow speed with an assumed repetition rate of the X-ray pulse train of 1.1 MHz and an assumed top hat beam profile of the X-ray laser with the dimension $10 \mu\text{m} \times 10 \mu\text{m}$. The minimum required flowspeed is calculated as follows:

$$\Delta x_{\text{travel}} \geq w \quad \Rightarrow \quad v_{\text{min}} \cdot \Delta\tau \geq w \quad \Rightarrow \quad v_{\text{min}} \geq \frac{w}{\Delta\tau} \Rightarrow v_{\text{min}} = 11 \text{ m s}^{-1}$$

With $w = 10 \mu\text{m}$ being the spatial width and $\Delta\tau = \frac{1}{f} = \frac{1}{1.1 \times 10^6 \text{ Hz}} \approx 9.09 \times 10^{-7} \text{ s}$ the characteristic time.

There are three laser wavelengths available for excitation: 355 nm, 400 nm and 515 nm. To evaluate which fits the experiment best, the concentration c of the solution for the three wavelengths for a desired optical density of $OD = 1$ and a liquid jet thickness of $d = 25 \mu\text{m}$ was calculated. The optical density is defined as $OD = \epsilon \cdot c \cdot d$ with $\epsilon(\lambda)$ the wavelength dependent extinction coefficient of $[\text{Fe}(\text{bipy})_3]^{2+}$ in water. The extinction coefficients can be estimated to be $\epsilon(355 \text{ nm}) = 12000 \text{ M}^{-1} \text{ cm}^{-1}$, $\epsilon(400 \text{ nm}) = 4000 \text{ M}^{-1} \text{ cm}^{-1}$, $\epsilon(515 \text{ nm}) = 8000 \text{ M}^{-1} \text{ cm}^{-1}$ as taken from Figure 11.

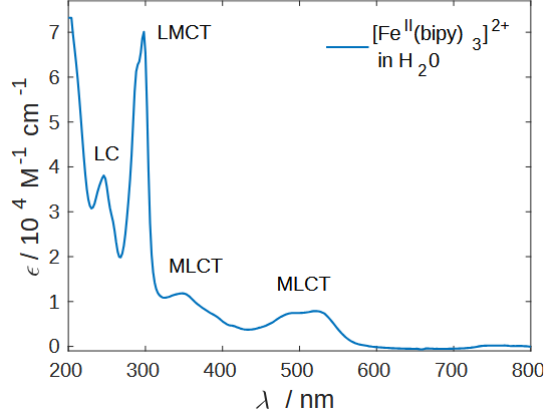


Figure 11: UV/VIS absorption spectrum of an aqueous $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ solution with the different charge-transfer transitions indicated. The wavelength-dependent absorption is used to estimate suitable concentrations and to compare the available excitation wavelengths [4].

The following concentrations were calculated:

$$c_{355} = 0.033 \text{ mol/L}, c_{400} = 0.100 \text{ mol/L}, c_{515} = 0.050 \text{ mol/L}$$

There are different pulse energies available for each excitation wavelength: $5 \mu\text{J}$ (355 nm), $15 \mu\text{J}$ (400 nm) and $50 \mu\text{J}$ (515 nm). With that in mind the wavelength of 400 nm was chosen, as it has a higher pulse energy, even though it's not ideal when looking at the low absorption rate. The extinction coefficient of the 515 nm is slightly better but with the low pulse energy it will be hard to see anything.

As we want to ensure that the X-ray beam always hits excited molecules we need to determine the diameter of the optical laser beam as well as the diameter of the X-ray beam correctly in relation to each other.

The optical laser beam profile is approximately gaussian-shaped and has an original diameter of 10 mm (FWHM). This collimated beam is focussed onto the liquid jet by a lens with a focal length of $f = 250 \text{ mm}$. From the Rayleigh criterion $d_{\text{Airy}} = 2.44 \cdot \frac{\lambda \cdot f}{D}$ we calculate $d_{\text{Airy}} = 24.4 \mu\text{m}$ with our chosen wavelength of 400 nm.

If the optical laser and the X-ray laser had the same diameter, the x-ray beam would also probe molecules that were excited only weakly or even not at all, as the optical laser has an approximately Gaussian beam shape. Ideally, the X-ray is positioned exactly in the centre of the ca. 10 times bigger optical laser beam. Thereby it can investigate a maximal volume of excited molecules. This also compensates for the fact that the optical laser jitters in the focus layer of the X-ray due to not being focussed there itself.

These estimates have consequences for our needed flow speed of the liquid jet which has to be 10 times faster to adjust to the new diameter of the optical laser.

Ideally, the number of absorbed optical laser photons should be less than or approximately equal to the number of sample molecules in the laser-excited volume. This reduces the likelihood that multiple photons hit the same molecule and excite it into states other than the one we want to observe while still optimizing the number of excited molecules.

To ensure the optical laser excites the sample efficiently, the laser intensity ideally is in a range where the number of optical laser photons is less than or approximately equal to the number of sample molecules in the laser-excited volume.

First we calculate the number of $[\text{Fe}(\text{bipy})_3]^{2+}$ molecules in the volume illuminated by the laser with a chosen beam size of $d_L = 150 \mu\text{m}$, a concentration of $c = 1.00 \times 10^{-4} \text{ mol/cm}^3$, a wavelength of $\lambda = 400 \text{ nm}$ and a jet size of $d_{\text{jet}} = 25 \mu\text{m}$. The particle density is calculated by:

$$\mathcal{N} = c \cdot N_A = 1.00 \times 10^{-4} \text{ mol/cm}^3 \cdot 6.022 \times 10^{23} \text{ mol}^{-1} = 6.022 \times 10^{19} \text{ molecules/cm}^3 \quad (3)$$

The illuminated cross-section area A of the Gaussian beam is $A_{\text{eff}} = 1.133 \cdot d_{\text{FWHM}}^2 = 1.133 \cdot (0.015 \text{ cm})^2 \approx 2.549 \times 10^{-4} \text{ cm}^2$. The effective volume is therefore $V_{\text{eff}} = A_{\text{eff}} \cdot d_{\text{jet}} = 2.549 \times 10^{-4} \text{ cm}^2 \cdot 25 \times 10^{-4} \text{ cm} = 6.373 \times 10^{-7} \text{ cm}^3 = 6.373 \times 10^{-10} \text{ L}$. The number of molecules is then calculated:

$$N_{\text{molecules, eff}} = \mathcal{N} \cdot V_{\text{eff}} = (6.022 \times 10^{19} \text{ cm}^{-3}) \cdot (6.373 \times 10^{-7} \text{ cm}^3) \approx 3.838 \times 10^{13} \text{ molecules} \quad (4)$$

Now to determine the pulse energy at which the number of photons per pulse is equal to the number of molecules just calculated:

$$N_{\text{photons}} = N_{\text{molecules}} \quad (5)$$

$$E_{\text{photon}} = \frac{h \cdot c}{\lambda} = \frac{(6,626 \times 10^{-34} \text{ J} \cdot \text{s}) \cdot (2,998 \times 10^8 \text{ m/s})}{400 \times 10^{-9} \text{ m}} \approx 4.966 \times 10^{-19} \text{ J} \quad (6)$$

$$E_{\text{pulse}} = N_{\text{molecules}} \cdot E_{\text{photon}} = (3.838 \times 10^{13}) \cdot (4.966 \times 10^{-19} \text{ J}) \approx 1.906 \times 10^{-5} \text{ J} \quad (7)$$

$$= 19.06 \mu\text{J} \quad (8)$$

For a wavelength of 400 nm, the equal-photon threshold is approximately 19.06 μJ for a spot diameter of 150 μm . This value is slightly above the available pulse energy of 15 μJ , but it is only an order-of-magnitude estimate because it assumes one absorbed photon per excited molecule and neglects saturation and the detailed excitation probability.

There are two X-ray crystal spectrometers available to use: a von Hamos spectrometer and a Johann spectrometer. As the von Hamos spectrometer can record the entire spectrum at once while also being immune to intensity variation, in contrast to the Johann spectrometer which needs to scan each wavelength separately and requires normalization to the accumulated incident intensity at each measurement point to measure the whole spectrum, it is better suited for this experiment.

In order to choose the correct crystal for the emission spectrometer the Bragg reflection angle needs to be considered which means we need to calculate the Bragg reflection angle for the Fe $K\beta_{1,3}$ emission line. From the FXE website [?] we preselect the Si(531) crystal and the Ge(531) crystal. The Bragg angle is calculated through:

$$\lambda = 2d_{hkl} \cdot \sin(\theta_B) \quad (9)$$

With $\lambda_m = 1.7566 \text{ \AA}$ for the Fe $K\beta_{1,3}$ main line ($E_m = 7058 \text{ eV}$), $\lambda_s = 1.7596 \text{ \AA}$ for the Fe $K\beta'$ satellite line ($E_s = 7046 \text{ eV}$), d calculated through the crystal planes (hkl) and the lattice constants $a_{\text{Si}} \approx 5.431 \text{ \AA}$ & $a_{\text{Ge}} \approx 5.658 \text{ \AA}$ to $d_{\text{Si}} = \frac{5.431}{\sqrt{35}} \approx 0.918 \text{ \AA}$ and $d_{\text{Ge}} = \frac{5.658}{\sqrt{35}} \approx 0.956 \text{ \AA}$.

For the Fe $K\beta_{1,3}$ main emission line the following Bragg angles were calculated:

$$\theta_{B_{\text{Si}}} = \arcsin\left(\frac{1.7566}{2 \cdot 0.9180}\right) \approx 73.09^\circ \quad (10)$$

$$\theta_{B_{\text{Ge}}} = \arcsin\left(\frac{1.7566}{2 \cdot 0.9564}\right) \approx 66.69^\circ \quad (11)$$

For the Fe $K\beta'$ satellite line the following Bragg angles were calculated:

$$\theta_{B_{\text{Si}}} = \arcsin\left(\frac{1.7596}{2 \cdot 0.9180}\right) \approx 73.42^\circ \quad (12)$$

$$\theta_{B_{\text{Ge}}} = \arcsin\left(\frac{1.7596}{2 \cdot 0.9564}\right) \approx 66.92^\circ \quad (13)$$

For a Bragg angle that is closer to 90° the spectral resolution is better. As Si(531) has a Bragg angle of 73.1° which is the highest of the two materials, it was chosen for the Von Hamos even though germanium has a higher atomic number than silicon and would have a higher reflection-efficiency. The higher Bragg angle of silicon makes a given energy difference produce a larger angular separation, so the improved spectral resolution is more important here than germanium's higher reflection efficiency.

4.2. Performing the Experiment with Data Acquisition

Using the Large Pixel Detector one can optimize the position of the liquid jet in the X-ray beam for maximum intensity of elastic scattering. The maximum intensity achieved with a pink beam was $3.8 \cdot 10^{15}$. The X-ray emission spectrometer was adjusted by setting the Bragg angle range to 74.49° to 72.03° on the crystal and 70.43° to 75.53° on the detector and orienting the detector perpendicular to the incoming dispersive spectrum (see Figure 12).

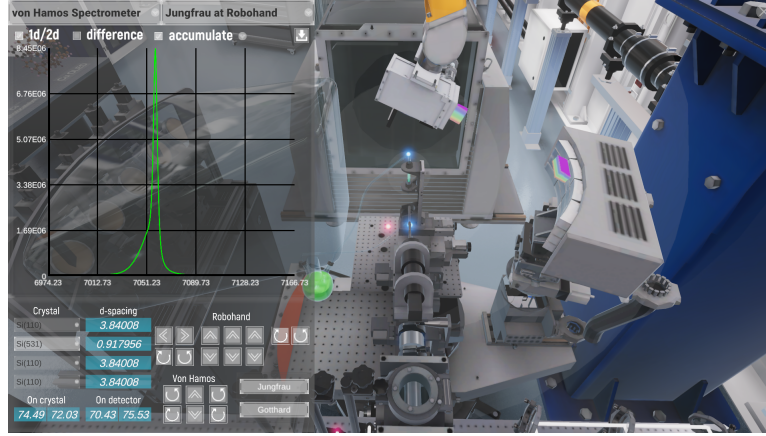


Figure 12: The Von Hamos spectrometer in its operating position, with the Jungfrau detector read-out shown alongside the instrument settings. This arrangement records the Fe $K\beta$ emission spectrum over a broad energy range in a single measurement.

The excitation laser was set to $\lambda = 400$ nm and $f = 0.02$ with a beam diameter of $d = 150 \mu\text{m}$ in the sample plane. The spatial overlap between laser and X-ray beam was adjusted so that the X-ray laser always overlapped with the optical laser even with the jitter of the optical laser. The corresponding beam profiles can be seen in Figure 15, while the laser beam profile can be seen in Figure 14 and the X-ray beam profile in Figure 13.

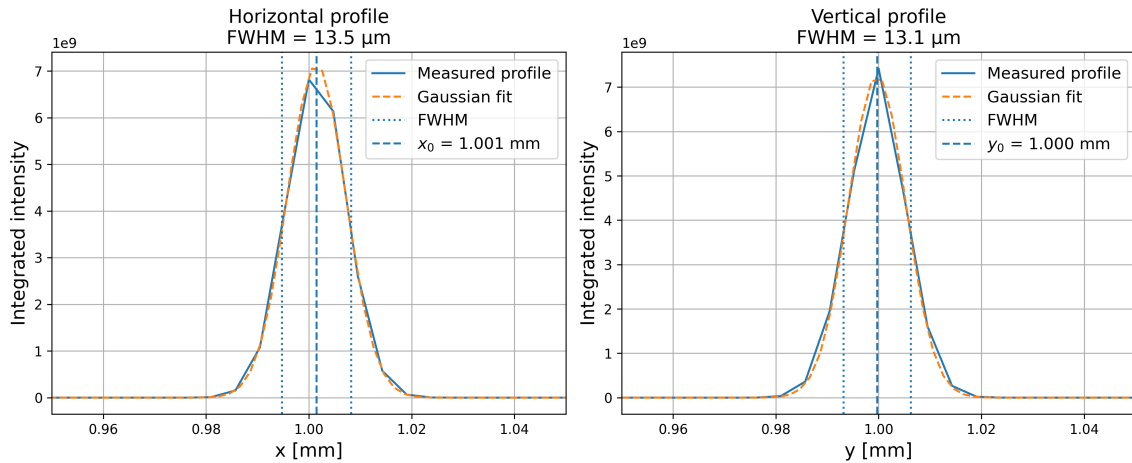


Figure 13: Spatial intensity profile of the focused X-ray beam in the sample plane. This measurement defines the probe-beam position and size at the liquid jet.

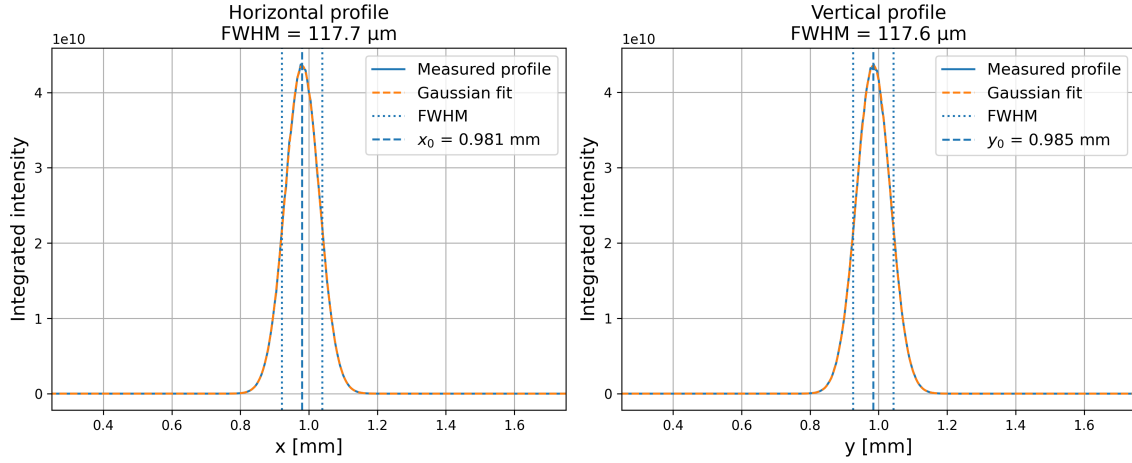


Figure 14: Spatial intensity profile of the optical excitation laser in the sample plane. Its broader profile is used to illuminate a larger region than the focused X-ray probe.

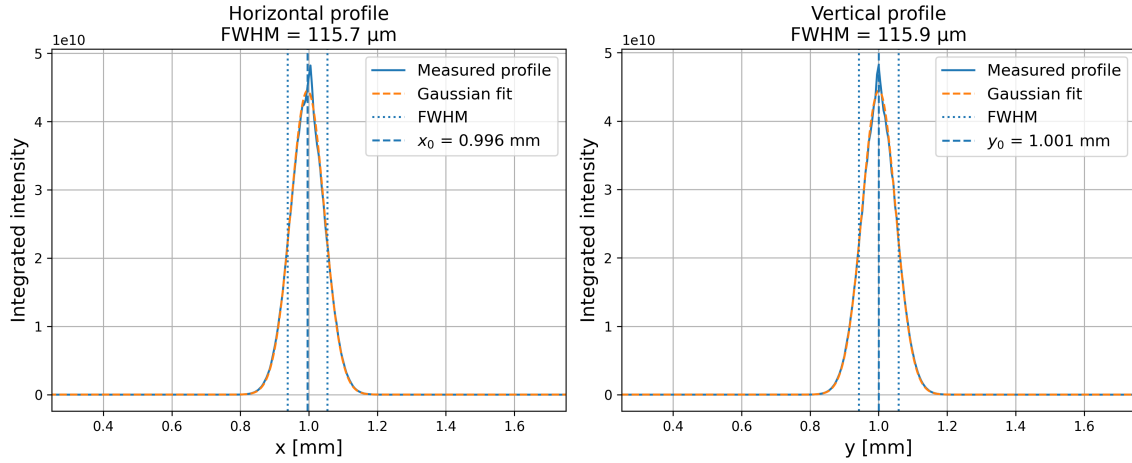


Figure 15: Superposition of the optical laser and X-ray beam profiles in the sample plane. The overlap demonstrates that the X-ray probe is positioned inside the laser-excited region despite the spatial jitter of the optical beam.

First the time delay of the laser was set to 100 ps. This time delay means the laser pulse definitely strikes the sample before the X-ray pulse arrives. XES spectra with the laser on and off were recorded, as well as a difference spectrum.

For the main measurement of the photophysical/photochemical reaction of $[\text{Fe}(\text{bipy})_3]^{2+}$ the range of delays was chosen to be -150 fs to 800 fs with a step size of 50 fs between -150 fs and 400 fs and a step size of 200 fs between 400 fs and 800 fs. For each delay ten difference spectra were recorded with an accumulation time of $60 \text{ s} \pm 2 \text{ s}$ each, which was tracked using a stopwatch.

5. Analysis and Interpretation

The analysis of the measured data is carried out using provided reference spectra resolved into the different spin states (see Figure 16).

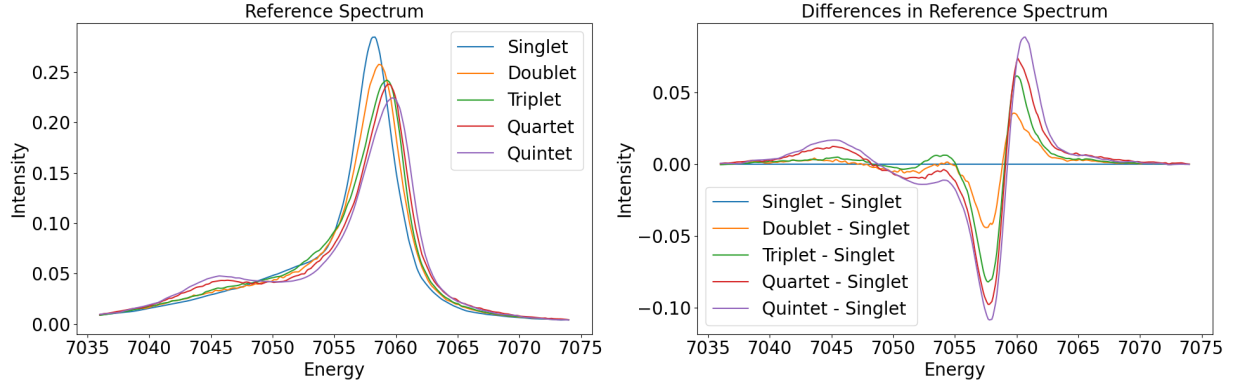


Figure 16: Reference X-ray emission spectra for the different spin states together with the corresponding reference difference spectra. These spectra provide the basis for identifying the spin-state contributions in the measured transient signal.

5.1. XES spectra at 100ps

When plotting the XES spectrum and the difference spectrum for a time delay of 100 ps, a clear transient signal is recognisable with the expected peaks from the reference spectrum and the XES spectrum shows similarities with the singlet spectrum from the reference (see Figure 17).

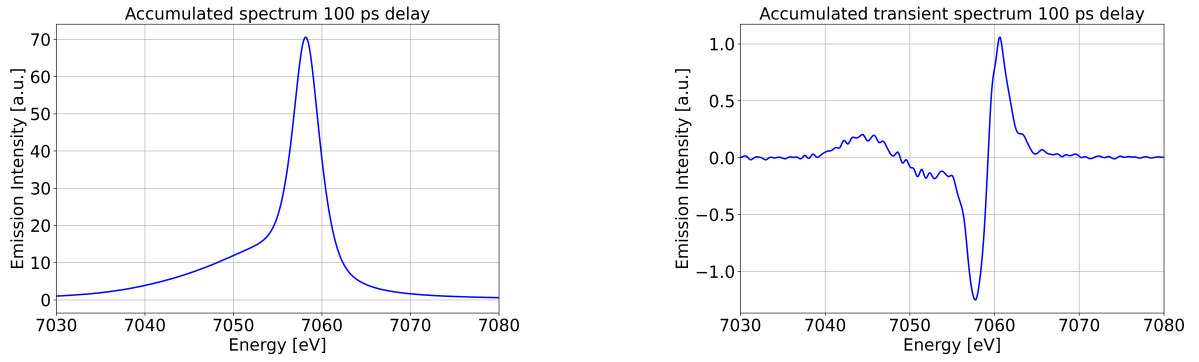


Figure 17: X-ray emission spectrum measured with the excitation laser on at a delay of 100 ps (left) and the corresponding laser-on minus laser-off difference spectrum (right). The comparison separates the dominant emission features from the transient changes caused by the optical excitation.

5.2. Difference spectra for different pump-probe delays

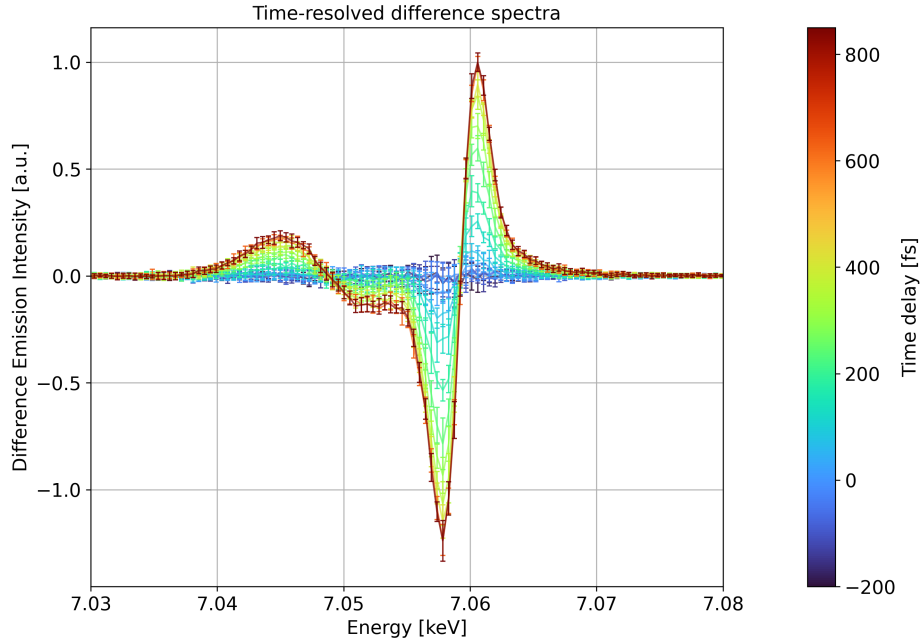


Figure 18: Transient X-ray emission spectra recorded at different pump-probe delays. Their evolving positive and negative features reveal the spectral changes associated with the population transfer between spin states.

From the time resolved difference spectra of the measurements (see Figure 18) we can see multiple developments in the transient signal over the different time delays: at an energy around 7045.0 eV the difference emission intensity grows (the intensity for the excited state is more intense than that of the ground state), at about 7053.7 eV a shoulder in intensity difference initially emerges and then flattens out again for time delays $\Delta t > 350$ fs. The two most prominent features of the transient signal are the minimum at 7057.8 eV and the maximum at 7060.5 eV which both match up with the difference spectrum provided.

Another thing to note is how little the change in spectrum between 600 fs and 800 fs is. The reaction has clearly landed on a more stable state, even though we can tell that it is still an excited state when comparing to the reference spectrum as it most closely matches the quintet-singlet transition denoted there. The comparison with the reference spectrum also tells us that there likely is a short-lived intermediate state as the shoulder we see resembles the triplet or quartet spectra.

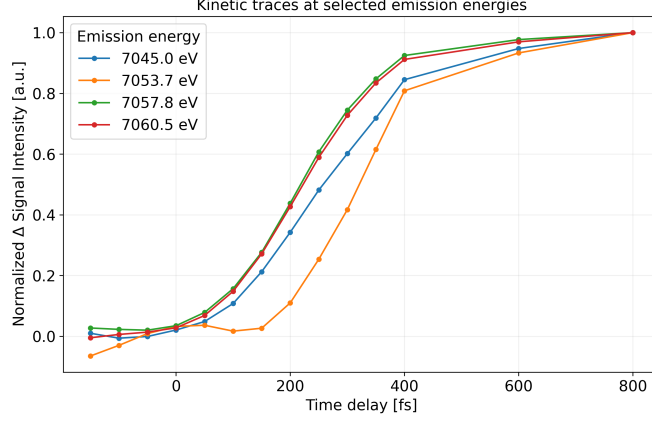


Figure 19: Kinetic traces of the transient signal at four selected emission energies as a function of pump-probe delay. Comparing these energies shows that the spectral features change systematically in time, confirming that the signal contains measurable dynamics rather than isolated statistical fluctuations and providing suitable traces for the subsequent analysis.

The kinetic traces at these energies (see Figure 19) show the time-dependent changes of the main positive, negative and shoulder-like spectral features. Their different temporal behaviour confirms that the transient signal contains evaluable dynamics and can be used to determine the rise time of the excited-state populations of $[\text{Fe}(\text{bipy})_3]^{2+}$ during the relaxation.

To determine the populations, the measured difference spectrum is modelled as a linear combination of the five reference spectra. The singlet spectrum $S_1(E)$ represents the ground state and is therefore used as the laser-off reference. The model for a measurement at delay t is

$$D(E, t) = A \left(\sum_{s=1}^5 f_s(t) S_s(E) - S_1(E) \right), \quad \sum_{s=1}^5 f_s(t) = 1. \quad (14)$$

Here, $f_s(t)$ is the population of spin multiplicity s and A is a common amplitude. The amplitude is not fitted independently for every delay but is first calibrated using the spectrum at $t = 800$ fs under the assumption that this spectrum is purely quintet:

$$A = \frac{\sum_E \frac{D(E, 800 \text{ fs}) [S_5(E) - S_1(E)]}{\sigma^2(E, 800 \text{ fs})}}{\sum_E \frac{[S_5(E) - S_1(E)]^2}{\sigma^2(E, 800 \text{ fs})}}. \quad (15)$$

This calibrated value is then kept fixed for all delays. For each spectrum, only the four excited-state populations f_2 , f_3 , f_4 and f_5 are optimised by minimising the weighted residual

$$\chi^2 = \sum_E \left[\frac{D(E, t) - D_{\text{fit}}(E, t)}{\sigma(E, t)} \right]^2, \quad (16)$$

with $0 \leq f_s \leq 1$ and $\sum_{s=2}^5 f_s \leq 1$. The singlet population is not an independent fit parameter, but is calculated from the normalisation condition:

$$f_1(t) = 1 - f_2(t) - f_3(t) - f_4(t) - f_5(t). \quad (17)$$

The resulting populations and their uncertainties are shown in Figure 20. To estimate the transition time of the singlet population, $f_1(t)$ is additionally fitted with a complementary error function:

$$f_1(t) = f_{\text{after}} + \frac{\Delta f}{2} \operatorname{erfc} \left(\frac{t - t_0}{\sqrt{2} \sigma_t} \right). \quad (18)$$

Here, t_0 is the midpoint of the population drop and σ_t is the standard deviation of the Gaussian temporal response. The corresponding full width at half maximum is $2\sqrt{2\ln 2}\sigma_t$.

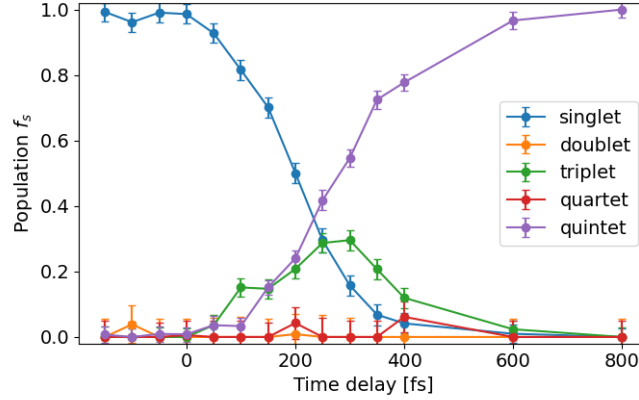


Figure 20: Fitted populations of the singlet, doublet, triplet, quartet and quintet reference states as a function of pump-probe delay. The transient triplet contribution and the subsequent growth of the quintet population support a relaxation pathway through an intermediate state.

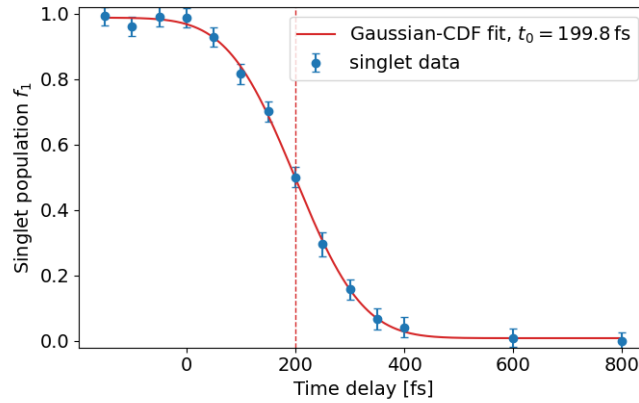


Figure 21: Singlet population extracted from the spectral decomposition together with the Gaussian temporal-transition fit. The fitted midpoint t_0 marks the effective temporal position of the excitation, while the fit width describes the finite temporal spread of the pump-probe response.

6. Discussion of the physical meaning

The population fit shows the conversion of the low-spin singlet ground state into excited spin states after excitation by the optical laser. The transient triplet population first increases and later decreases again, while the quintet population grows and becomes dominant at long delays. This temporal ordering is consistent with a short-lived triplet intermediate that transfers population to the long-lived quintet state. The data therefore support a predominantly sequential relaxation pathway rather than a direct transition to the quintet state alone. Small doublet and quartet contributions occur only at some delays and remain much smaller than the main populations. They are likely caused by correlations between the reference spectra and statistical fluctuations, so they do not provide sufficient evidence for additional physical states.

The fitted midpoint of the singlet-population drop is $t_0 = 199.75 \pm 5.79$ fs. In the context of the Gaussian temporal statistics of the laser pulses, we interpret this midpoint as the effective start of the excitation. The Gaussian model accounts for the fact that the ultrashort pump and probe pulses have

a finite temporal width rather than interacting at one perfectly defined instant. The fitted transition width is $\sigma_t = 97.63 \pm 7.63$ fs, corresponding to $\text{FWHM} = 229.91 \pm 17.96$ fs. Thus, t_0 marks the temporal position of the excitation in our measurement, while the width describes its temporal spread.

At long delays, the population is almost entirely quintet, consistent with the long-lived high-spin state. The unity quintet population at 800 fs should be interpreted with care, since this spectrum was used to calibrate the common amplitude under the pure-quintet assumption.

6.1. Error analysis

The error bars of the difference-spectrum measurements arise from statistical fluctuations in the data, which are the only source of error modeled in VLab. They are about two orders of magnitude smaller than the largest peak in arbitrary units. To reduce the error bars by half, the difference spectra would have had to be accumulated for four times as long. This was not feasible during the measurements, especially because the program had already crashed twice. Therefore, we preferred collecting data for all time delays with somewhat larger uncertainties rather than risking incomplete measurements.

7. Conclusion and outlook

During the experiment the X-ray beam was successfully characterised and focussed to a usable point with a FWHM of $13.4 \mu\text{m}$. The following settings for the pump-laser were determined:

Laser power	Repetition rate	Pulse energy	Wavelength	Full width, vertical	Full width, horizontal
27.5 W	550 000 Hz	5.0×10^{-5} J	400 nm	$150 \mu\text{m}$	$150 \mu\text{m}$

Then a successful series of pump-probe-measurements was carried out in which we can see the occurrence of an ultrafast intermediate state can be seen. The identification was carried out using a provided reference spectrum with which it was possible to fit time delay-dependant population curves resolved after the different spin states present during the relaxation process.

The experiment successfully provided insight into the work on a real beamline.

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Usage of AI In this work, AI tools were used as supportive aids for proofreading and language refinement, as well as for assistance with LaTeX formatting and code development. Specifically, Gemini was used for text proofreading and Copilot for LaTeX and code-related help. The authors have reviewed all AI-suggested edits. The scientific content and final conclusions were independently developed by the authors.

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Appendix A Additional Plots

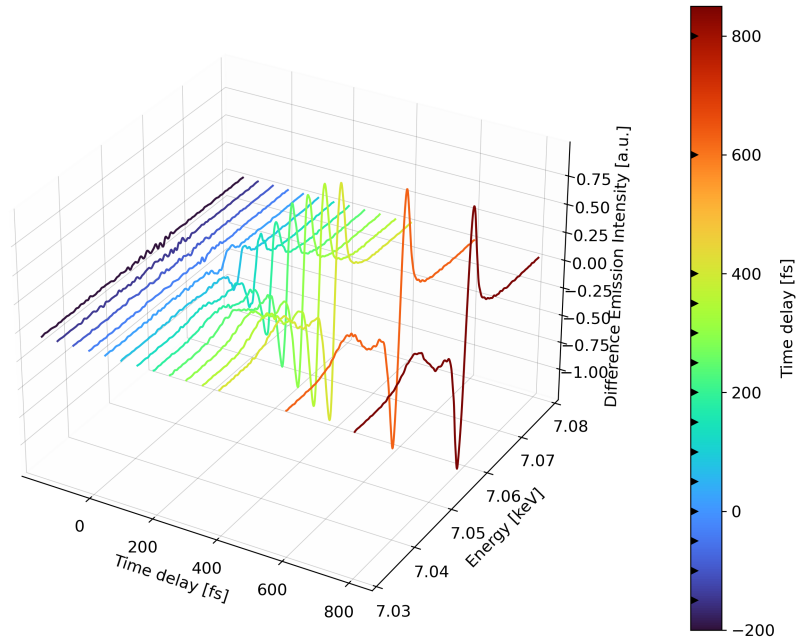


Figure 22: Three-dimensional representation of the time-resolved transient X-ray emission spectra. The additional view highlights how the spectral features evolve continuously with the pump-probe delay.

Appendix B Code